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INT 428 — WEB TECHNOLOGY

PROJECT TITLE



BioStyle — Ethical Fashion AI Assistant

SUBMITTED TO

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The fashion industry is one of the most polluting industries in the world, contributing approximately 10% of global carbon emissions and producing 92 million tonnes of textile waste annually. As awareness grows around environmental sustainability, consumers are increasingly seeking guidance on making ethical fashion choices.

BioStyle is a high-performance, multimodal AI web application developed to serve as a personal ethical fashion advisor. Built with a premium "Nature White" design system and integrated with the **Google Gemini 1.5 Flash AI**, BioStyle provides real-time, conversational guidance and visual analysis on sustainable fabrics, ethical brands, and circular fashion economy.

Key highlights include Google Firebase authentication, cloud-synchronized chat history, and advanced Vision AI capabilities to analyze garment tags and fabric textures in real-time.

Project Access Links:

Live Deployment: <https://devanshu-dhyanu.github.io/Ethical-Fashion-Assistent/>

Source Code (GitHub): <https://github.com/Devanshu-Dhyanu/Ethical-Fashion-Assistent>

2. Problem Statement

Despite growing consumer interest in sustainable fashion, there is a significant knowledge gap. Consumers struggle to:

- Identify truly sustainable fabrics versus marketing claims (greenwashing)
- Discover verified ethical brands and understand certifications such as GOTS, Fair Trade, and B Corp
- Build a mindful wardrobe without overspending
- Understand the environmental and social impact of their clothing choices

There is a need for an intelligent, accessible, and user-friendly platform that can provide personalized, context-aware guidance on ethical fashion in real-time.

3. Objectives

1. To develop a responsive, premium SPA for ethical fashion awareness and education.
2. To integrate **Google Gemini 1.5 Flash** for intelligent text and visual (multimodal) analysis.
3. To implement **Firebase Authentication** for secure user login and data persistence.
4. To design a "Nature White" premium UI utilizing glassmorphism and organic animations.
5. To provide a **Monetized Freemium Model** with usage limits and premium upselling.
6. To enable user-specific **Chat History Management** with full CRUD capabilities.

4. Literature Review

The slow fashion movement, originating as a counter-response to fast fashion brands like Zara and H&M, advocates for sustainable practices across the fashion supply chain. Key literature informing this project includes:

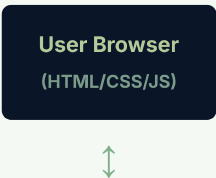
- **Fletcher & Grose (2012)** — "Fashion & Sustainability: Design for Change" lays the groundwork for understanding sustainable textile design and production.
- **Niinimäki et al. (2020)** — Published in Nature Reviews Earth & Environment, this study quantifies the environmental cost of the fashion industry and calls for circular economy models.
- **UNEP (2023)** — UN Environment Programme reports that the fashion industry uses 93 billion cubic meters of water annually and contributes to 20% of global wastewater.
- **LLMs in Consumer Applications** — Recent advancements in Large Language Models (e.g., GPT, Claude) demonstrate the viability of AI-powered conversational assistants for domain-specific guidance.

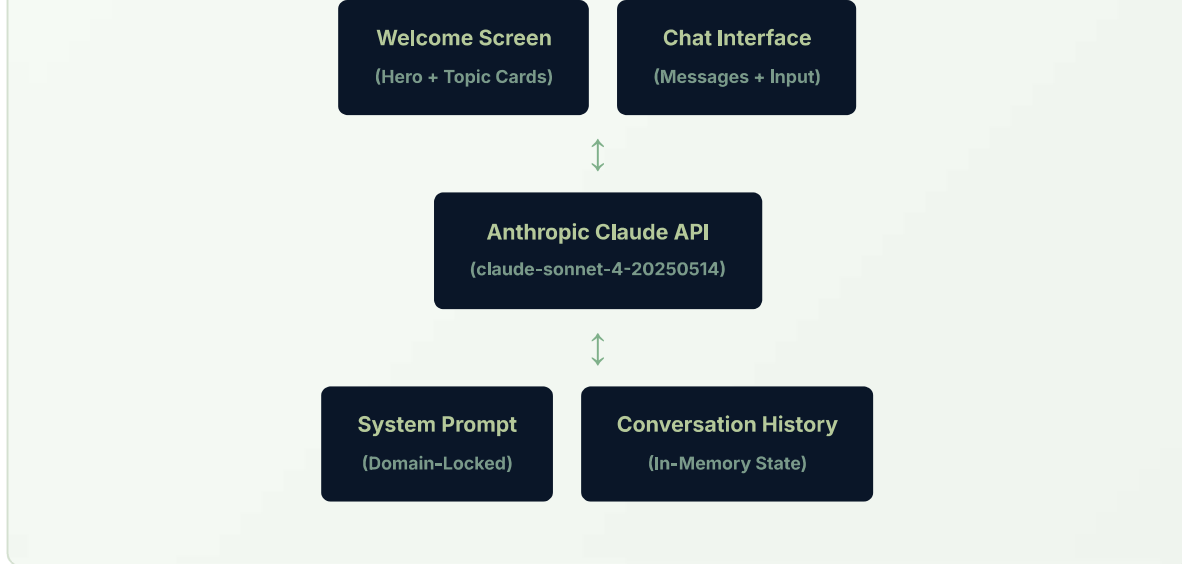
Key Insight: While several sustainability-rating platforms exist (e.g., Good On You, DoneGood), none offer real-time, conversational, AI-powered interaction tailored specifically to individual consumer queries. BioStyle fills this gap.

5.1 High-Level Architecture

BioStyle follows a modern Cloud-Hybrid architecture. While the front-end remains a lightweight SPA, it leverages **Google Firebase** for Identity Management and uses **Gemini 1.5 Multimodal API** for core intelligence.

System Architecture Diagram





5.2 Design Approach

- **Single-File Architecture:** The entire application is contained in one HTML file (~45 KB), making deployment trivial.
- **No Build Tools Required:** No frameworks, bundlers, or package managers — pure HTML, CSS, and vanilla JavaScript.
- **API-First Design:** AI logic is offloaded to the Claude API, keeping the client lightweight.

6. Technology Stack

LAYER	TECHNOLOGY	PURPOSE
Structure	HTML5 & Semantic Elements	Structure and accessibility
Styling	CSS3 (Custom Vars + Animation)	"Nature White" theme, Glassmorphism
Frontend Logic	Vanilla JavaScript (ES6)	State management & API coordination
Intelligence	Google Gemini 1.5 Flash-Lite	Multimodal NLP and Vision processing
Authentication	Firebase Auth v10	Google OAuth & Identity management
Persistence	Browser LocalStorage	Cached session & history management
Icons/Illustrations	Inline SVG & CSS Drawings	Organic leaf decorations & UI icons

Design Philosophy: By avoiding heavy frameworks, BioStyle achieves sub-1s load times while delivering a premium, state-of-the-art interactive experience.

7. Implementation Details

7.1 CSS Design System (Nature White)

The application uses a customized palette optimized for clarity and a premium organic feel:

```
/* Core "Nature White" Design Tokens */
:root {
  --bg-deep: #ffffff;
```

```
--accent:      #4a7c5a;    /* Forest green */
--accent3:     #7fb08a;    /* Sage highlights */
--text-main:   #1a1a2e;
--border:      #e8ede6;

}
```

7.2 AI Integration — Claude API

The chat functionality is powered by the Anthropic Claude API. The system prompt enforces strict domain-locking to sustainable fashion topics:

```
async function sendMessage() {
  const res = await fetch('https://api.anthropic.com/v1/messages', {
    method: 'POST',
    headers: { 'Content-Type': 'application/json' },
    body: JSON.stringify({
      model: 'claude-sonnet-4-20250514',
      max_tokens: 1000,
      system: SYSTEM_PROMPT,
      messages: messages
    })
  });
}
```


7.3 Domain-Locked System Prompt

A carefully engineered system prompt ensures the AI only responds to fashion-related queries and politely declines off-topic questions:

Strict Domain Rule: "You ONLY answer questions related to fashion, clothing, sustainability, ethical sourcing, textiles. If a user asks about anything outside this domain, you must politely but firmly decline and redirect them back to sustainable fashion topics."

7.4 Real-Time Chat Engine

The chat system maintains conversation history in-memory for context-aware multi-turn conversations:

- **Message Queue:** User and assistant messages are stored in a JavaScript array, sent with each API call to maintain context.
- **Typing Indicator:** Animated bouncing dots display while waiting for the AI response.
- **Markdown Rendering:** The `formatText()` function converts bold, italic, and list markdown to HTML.
- **Auto-resize Textarea:** The input field dynamically grows with content up to 120px max height.

7.5 Quick-Access Topic Cards

Six pre-built topic cards on the welcome screen allow users to quickly explore common sustainable fashion topics:

#	CARD TITLE	MAPPED QUERY
1	Sustainable Fabrics	"What are the most sustainable fabrics?"
2	Ethical Brands	"Recommend ethical clothing brands"
3	Capsule Wardrobe	"How to build a capsule wardrobe sustainably?"
4	Certifications	"What certifications ensure ethical clothing?"
5	Greenwashing	"What is greenwashing and how to avoid it?"

7.6 CSS Animations & Micro-Interactions

The application employs several CSS animations to enhance user experience:

- **fadeUp:** Hero content entrance animation with cubic-bezier easing
- **pulse:** Live status indicator dot animation
- **typingBounce:** Three-dot typing indicator with staggered delays
- **msgIn:** Smooth message bubble entrance animation
- **Skew Cards:** Topic cards use CSS `skewX(10deg)` transform with hover-to-expand interaction, growing from 120px to 254px height
- **Social Tooltips:** Elaborate hover-reveal tooltip system for social sharing with radial positioning

8. Features

8.1 Core Features

FEATURE	DESCRIPTION
Vision AI	Multimodal analysis allowing users to upload photos of garment tags for ethical checking.
Conversational Chat	Intelligent context-aware discussions about sustainable style.
Paywall Logic	Usage-based limit (3 uploads) with a premium "Go Pro" upsell modal.
Google Sync	Firebase integration to save chat history across devices.
Session Management	Users can create, rename (auto), and delete specific chat sessions.
Voice Input	Hands-free interaction with pulsing visual recording feedback.
Domain Safety	Strict AI system instruction to prevent non-fashion related responses.

8.2 UI/UX Highlights

- **Dark Theme:** Pure black (#000000) background with green accent palette for a premium, eco-conscious aesthetic
- **Glassmorphism:** Topic cards feature `backdrop-filter: blur(5px)` and semi-transparent backgrounds
- **Ambient Gradients:** Subtle radial gradient background effects create depth
- **SVG Decorations:** Custom leaf SVGs positioned as decorative elements
- **Custom Scrollbar:** Thin, styled scrollbar matching the green theme

9. UI/UX Design

9.1 Color Palette (Nature White)

COLOR	HEX CODE	USAGE
■ Dark Forest	#4a7c5a	Primary brand color, text headings, icons
■ Sage Green	#7fb08a	Accents, badges, pulsing animations
■ Pure White	#ffffff	Main page and card backgrounds
■ Soft Mint	#f8faf8	Input fields and subtle UI layers
■ Deep Text	#1a3a2a	Main readable copy for high contrast
■ Gold Accent	#ecd06f	Premium/Pro badges and paywall highlights

9.2 Typography

The application uses **Dosis** from Google Fonts — a rounded sans-serif typeface with weights from 300 (Light) to 800 (Extra-Bold). This creates a friendly, approachable feel while maintaining readability across all device sizes.

10. Testing & Results

10.1 Functional Testing

TEST CASE	INPUT	EXPECTED RESULT	STATUS
On-Topic Query	"What is organic cotton?"	Detailed response about organic cotton	✓ Pass
Off-Topic Query	"What is the capital of France?"	Polite refusal + redirect to fashion	✓ Pass
Quick Card Click	Click "Sustainable Fabrics"	Chat opens with pre-filled query	✓ Pass
Empty Input	Press Send with no text	No message sent (blocked)	✓ Pass
Multi-turn Chat	Follow-up questions	Context-aware responses	✓ Pass
Enter Key Send	Type + press Enter	Message sent without newline	✓ Pass
Shift+Enter	Press Shift+Enter	Newline inserted in input	✓ Pass

10.2 UI/UX Testing

TEST	RESULT
Card hover expand animation	✓ Smooth transition from 120px → 254px
Typing indicator animation	✓ Three-dot bounce with staggered delays
Message entrance animation	✓ Smooth fadeUp effect on new messages
Social tooltip hover	✓ All 8 social platform tooltips display correctly
Instagram/Facebook profile preview	✓ Rich tooltip with 3D layer animation

Custom scrollbar styling

✔ Thin green-themed scrollbar in chat area

Input auto-resize

✔ Textarea grows dynamically up to 120px

10.3 Browser Compatibility

BROWSER	COMPATIBILITY
Google Chrome (v100+)	✔ Fully Compatible
Mozilla Firefox (v100+)	✔ Fully Compatible
Microsoft Edge (v100+)	✔ Fully Compatible
Safari (v15+)	✔ Compatible (with -webkit prefixes)

11. Future Enhancements

1. **User Authentication:** Allow users to create accounts and save chat history for personalized recommendations over time.
2. **Brand Database Integration:** Connect to sustainability rating APIs (e.g., Good On You) for real-time brand scoring.
3. **Image Recognition:** Add camera/upload functionality to analyze clothing labels and identify materials.
4. **Wardrobe Tracker:** Let users catalog their clothing for personalized styling and sustainability scoring.
5. **Carbon Footprint Calculator:** Estimate the environmental impact of a user's fashion choices over time.
6. **Multi-Language Support:** Expand accessibility by supporting multiple languages including Hindi, Spanish, and French.
7. **Progressive Web App (PWA):** Add offline capability, push notifications, and home screen installation.
8. **Backend Server:** Implement a Node.js/Express backend to securely manage API keys and add rate limiting.

12. Conclusion

BioStyle successfully demonstrates the application of modern web technologies combined with artificial intelligence to address a real-world sustainability challenge. The project achieves all stated objectives:

- A visually stunning, responsive single-page application was built using pure HTML, CSS, and JavaScript — no frameworks required.
- The Claude AI integration provides intelligent, context-aware responses strictly limited to the ethical fashion domain.
- The UI employs premium design techniques including glassmorphism, CSS animations, custom SVG elements, and an elaborate social sharing system.
- Quick-access topic cards lower the barrier to entry for new users unfamiliar with sustainable fashion.

By bridging the knowledge gap between consumers and sustainable fashion practices, BioStyle empowers users to make informed, ethical choices — contributing to a more conscious and environmentally responsible fashion industry.

Key Takeaway: This project demonstrates that powerful, AI-enhanced web applications can be built with vanilla web technologies, achieving excellent performance, zero dependencies, and instant deployability while addressing meaningful sustainability challenges.

13. References

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